Confidence-Weighted Pairs Trading via DCC-EGARCH *

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1 Introduction

A pairs trade exploits long-run economic equilibria by taking offsetting positions in assets when temporary price deviations occur. Three design choices govern such a strategy. The first is which pairs to trade, the second is when to enter and exit, and the third is how much capital each pair receives. Holding the pairs fixed, does optimizing the entry and exit thresholds pair by pair and weighting the pairs by a confidence score improve the risk-adjusted return of a pairs trading portfolio?

We backtest cointegration-based pairs trading strategies on a pool of US equities and exchange-traded funds (ETFs), over an evaluation window running from August 2021 to December 2023. Hedge ratios are time-varying and estimated by the Kalman filter of Kalman (1960). Spread volatility is modeled by the exponential generalized autoregressive conditional heteroskedasticity (EGARCH) model of Nelson (1991) within the dynamic conditional correlation (DCC) framework of Engle (2002). We optimize the entry and exit thresholds, and propose a confidence-weighting scheme based on the rolling cointegration statistic and forecast volatility. Net of transaction costs, optimizing the thresholds alone lowers the Sharpe ratio, while combining the optimized thresholds with the confidence weights delivers the highest total return, the highest Sharpe ratio and the lowest maximum drawdown of the three strategies.

Section 2 describes the data and forms the pairs, Section 3 models the spread volatility, Section 4 defines the trading signal and optimizes its thresholds, Section 5 sets out the weighting scheme, Section 6 reports the backtest and Section 7 concludes.[1]

2 Pairs Formation

2.1 Data

The asset pool is 45 US equities and ETFs of large market capitalization and high liquidity, listed in Table 1. It spans commodities, technology, healthcare, banking, energy, consumer staples and real estate. Daily closing prices are taken from January 2012 to December 2023, and names without a complete price history over that period are excluded. We apply an 80-20 train-test split, so the training set covers January 2012 to 5 August 2021 and the testing set covers 6 August 2021 to December 2023.

2.2 Cointegration

For two price series x_t and y_t, both integrated of order one, define the spread

$$s_t = y_t - (\beta_t x_t + \alpha_t) \tag{1}$$

*This is a class project for the Chinese University of Hong Kong's FINA 4380 course, completed by a group of four members.

[1] Code and replication data are available at *GitHub*.

GDX	GDXJ	GLD	AAPL	GOOGL	AMD
NVDA	CSCO	ORCL	TTWO	EA	HYG
LQD	JNK	SLV	USLV	SIVR	USO
QQQ	SPY	VOO	VDE	VTI	VDC
KXI	IBB	VHT	VNQ	IYR	MSFT
PG	TMF	UPRO	WFC	JPM	GS
CVX	XOM	INTC	COST	WMT	T
VZ	CMCSA	AMZN			

Table 1: Asset pool screened for pairs formation.

and call the series cointegrated when s_t is (weakly) stationary for constant $\alpha_t = \alpha$ and $\beta_t = \beta$. We test for cointegration using the two-step procedure of Engle and Granger (1987), and retain a pair when the null of no cointegration is rejected at the 5% level. Figure 1 shows the resulting p-values for the $\binom{45}{2}$ candidates in both regression directions. Of these, 73 pairs are significant and are carried forward, spanning 38 distinct tickers.

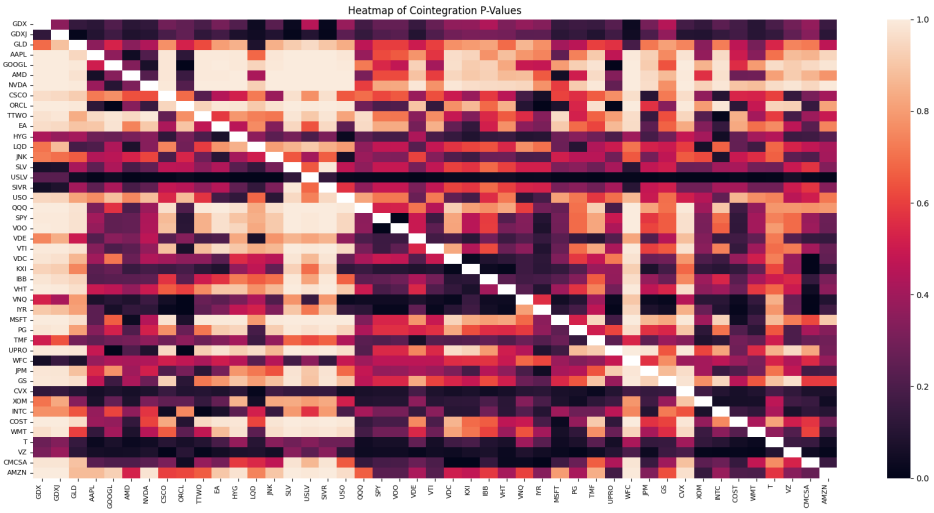


Figure 1: Engle-Granger p-values for every candidate pair on the training sample. Darker cells indicate stronger evidence of cointegration.

2.3 Time-varying Hedge Ratio

Market conditions and relative volatility move over time, so a hedge ratio estimated once by least squares will drift out of date. We therefore treat (α_t, β_t) in equation (1) as a latent state following a random walk, and estimate it with the Kalman filter of Kalman (1960). Writing the state as $\boldsymbol{\phi}_t = (\alpha_t, \beta_t)^\top$,

$$\begin{aligned} \boldsymbol{\phi}_t &= \boldsymbol{\phi}_{t-1} + \boldsymbol{\omega}_t, \qquad \boldsymbol{\omega}_t \sim N(0, \delta I), \\ y_t &= (1, x_t)\, \boldsymbol{\phi}_t + \nu_t, \qquad \nu_t \sim N(0, \sigma_\nu^2), \end{aligned} \tag{2}$$

with the state noise scale set to $\delta = 10^{-3}$ and the observation variance to $\sigma_\nu^2 = 2$. We pass each price series through a univariate Kalman smoother before running the regression to remove

transient noise. Then, we evaluate the spread in equation (1) at the filtered state.

2.4 Half-Life of Mean Reversion

The rolling window used later to compute the mean of each spread in its z-score is set by the speed at which that spread reverts. Regressing the spread increment on its lagged level,

$$\Delta s_t = \kappa\, s_{t-1} + \epsilon_t, \qquad H = -\frac{\ln 2}{\kappa}, \tag{3}$$

yields the half-life H, the number of days over which a deviation is expected to halve. Each pair uses its own H, rounded to whole days, as its window length.

3 Spread Volatility

We test each spread for normality with the Shapiro–Wilk test of Shapiro and Wilk (1965) and the Jarque–Bera test of Jarque and Bera (1980), which compares the sample skewness and kurtosis with those of the Gaussian. In the training sample, both tests reject normality at the 5% level for every pair, and the quantile-quantile (Q-Q) plots of the first 60 pairs in the Appendix show heavier tails than the Gaussian for most spreads. We therefore adopt a Student's t conditional distribution throughout.

3.1 Exponential GARCH

Engle (1982) introduced the autoregressive conditional heteroskedasticity (ARCH) model. The exponential form of Nelson (1991) is used here for two reasons. It admits an asymmetric response, so that a negative shock at $t-1$ raises the variance at t by more than a positive shock of the same size, which is the leverage effect. It also models the logarithm of the variance, so that positivity holds for any parameter values.

$$\begin{aligned} \varepsilon_t &= h_t \eta_t, \qquad \eta_t \sim t_\nu(0,1) \\ \ln h_t^2 &= \omega_h + \sum_{i=1}^{p} \Big(a_i \big(|\eta_{t-i}| - \mathbb{E}|\eta_{t-i}| \big) + \lambda_i \eta_{t-i} \Big) + \sum_{j=1}^{q} b_j \ln h_{t-j}^2 \end{aligned} \tag{4}$$

where ε_t is the spread net of its conditional mean, $t_\nu(0,1)$ the standardized Student's t distribution with ν degrees of freedom, h_t the conditional volatility, ω_h the constant, a_i the ARCH coefficients, b_j the GARCH coefficients and λ_i the leverage coefficients.

The choices of p and q are screened over the set $\{1,2,3,4,5\}$, evaluated by the Akaike and Bayesian information criteria, whose mean and median across pairs are shown in Figure 2. The Bayesian criterion is minimized at $p=3$ and $q=1$ on both the mean and the median, and the Akaike criterion changes little beyond that order.

3.2 Dynamic Conditional Correlation

The 73 spreads are not independent, so we estimate their volatilities jointly under the DCC model of Engle (2002). Let $\mathbf{s}_t$ denote the $n = 73$ spreads at time t and $\boldsymbol{\mu}$ their conditional means. Standardizing by the diagonal matrix of conditional volatilities yields

$$\mathbf{v}_t = D_t^{-1}(\mathbf{s}_t - \boldsymbol{\mu}), \qquad D_t = \operatorname{diag}(h_{1,t}, h_{2,t}, \ldots, h_{n,t}), \tag{5}$$

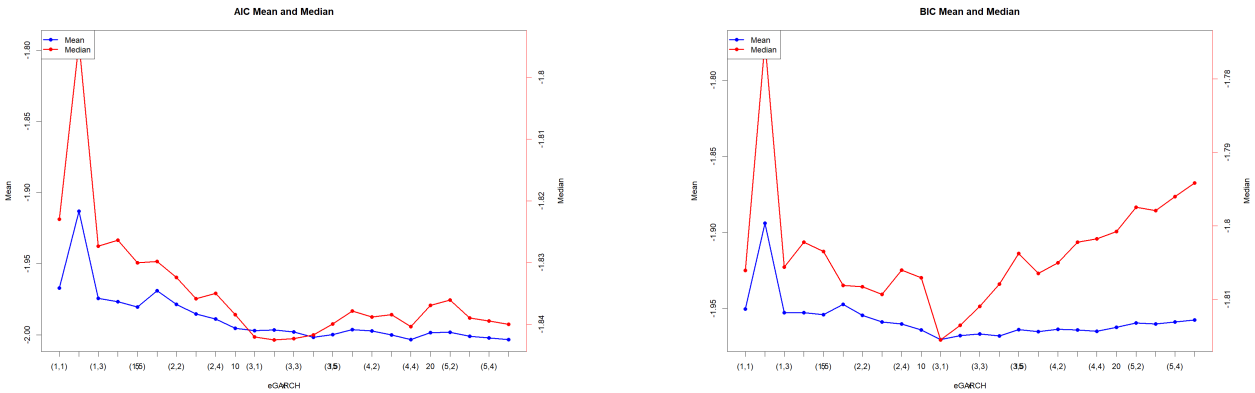


Figure 2: Akaike (left) and Bayesian (right) information criteria by EGARCH order (p, q).

and the quasi-correlation matrix evolves as

$$Q_t = (1 - \theta_1 - \theta_2)\bar{Q} + \theta_1 \mathbf{v}_{t-1}\mathbf{v}_{t-1}^\top + \theta_2 Q_{t-1}, \tag{6}$$

where θ_1 measures the response to the lagged standardized residuals, $\theta_1 + \theta_2$ the persistence, and $\bar{Q}$ is the unconditional level of Q_t. Denoting by T the number of training observations and by $T_0 = \lceil T/3 \rceil$ the end of the burn-in period, which is discarded so that the starting value does not contaminate the average,

$$\bar{Q} = \frac{1}{T - T_0 + 1} \sum_{t=T_0}^{T} \mathbf{v}_{t-1}\mathbf{v}_{t-1}^\top. \tag{7}$$

Fitting the model to the training spreads with multivariate Student's t innovations yields $\hat{\theta}_1 = 0.123$ and $\hat{\theta}_2 = 0.672$, a persistence of $\hat{\theta}_1 + \hat{\theta}_2 = 0.795$. Over the testing sample, the fitted model is iterated forward and the conditional volatility of each spread is extracted for the trading signal at every date.

4 Trading Signal

4.1 Definition

Each spread is standardized with $\bar{s}_{i,t}$ the rolling mean of spread i over its half-life window and $h_{i,t}$ the conditional volatility,

$$z_{i,t} = \frac{s_{i,t} - \bar{s}_{i,t}}{h_{i,t}}. \tag{8}$$

A position is opened when the z-score crosses an entry threshold and closed when it crosses the corresponding exit threshold, so that four parameters describe the rule. These are the long entry threshold LET, the long exit threshold LETX, the short entry threshold SET and the short exit threshold SETX. Crossings are strict, in the sense that the threshold must lie between consecutive observations,

$$\begin{aligned} \text{long entry} \quad & z_{i,t} < \text{LET}_i < z_{i,t-1}, & \qquad \text{long exit} \quad & z_{i,t-1} < \text{LETX}_i < z_{i,t}, \\ \text{short entry} \quad & z_{i,t-1} < \text{SET}_i < z_{i,t}, & \qquad \text{short exit} \quad & z_{i,t} < \text{SETX}_i < z_{i,t-1}. \end{aligned}$$

4.2 Threshold Optimization

Two objectives are considered: profit and Sharpe ratio. In both cases, the four thresholds are optimized by the Nelder-Mead simplex separately for each pair. Figure 3 reports the distribution of the fitted thresholds and Table 2 reports their summary statistics under both objectives. Under the profit and Sharpe ratio objectives, the median LET is -0.69 and -0.71, the median SET is 0.68 and 0.69, and the median LETX and SETX are 0.05 and -0.05 under both.

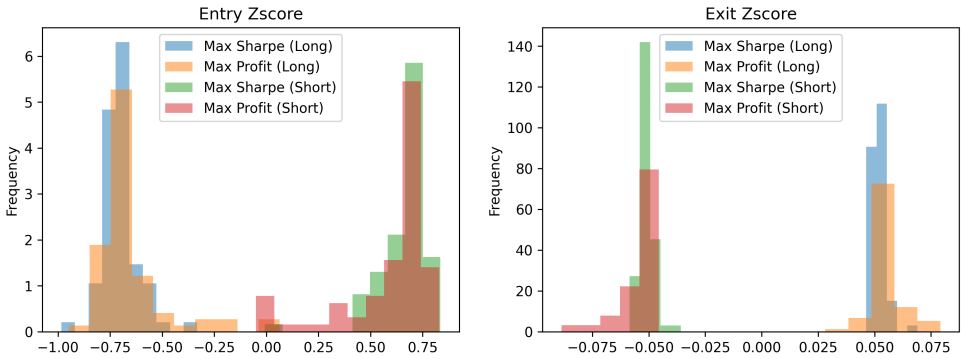


Figure 3: Histograms (density scale) of the optimized entry (left) and exit (right) thresholds under both objectives.

	Objective	LET	LETX	SET	SETX
Panel A. Maximum profit					
Mean	3.438	−0.647	0.054	0.588	−0.054
Standard deviation	1.636	0.177	0.007	0.221	0.008
Minimum	1.213	−0.949	0.028	−0.050	−0.089
25th percentile	2.415	−0.730	0.051	0.549	−0.055
Median	2.897	−0.693	0.052	0.680	−0.052
75th percentile	3.530	−0.641	0.056	0.709	−0.050
Maximum	8.043	0.064	0.079	0.830	−0.046
Panel B. Maximum Sharpe ratio					
Mean	2.034	−0.701	0.052	0.659	−0.051
Standard deviation	0.502	0.088	0.003	0.117	0.003
Minimum	0.453	−0.984	0.046	−0.007	−0.058
25th percentile	1.810	−0.747	0.050	0.620	−0.053
Median	2.021	−0.707	0.051	0.695	−0.051
75th percentile	2.359	−0.665	0.053	0.725	−0.050
Maximum	3.236	−0.332	0.069	0.835	−0.036

Table 2: Summary statistics of the optimized thresholds and of the objective across pairs under the maximum profit (Panel A) and maximum Sharpe ratio (Panel B) objectives.

5 Portfolio Weighting

5.1 Confidence Score

Since equal weighting ignores the fact that the 73 pairs differ in how reliably they revert, we propose a confidence score based on the rolling cointegration statistics and forecast volatility. The first is the

strength of the cointegration relationship over a trailing window of $r = 250$ trading days, measured either by the absolute test statistic $\mathrm{Coint}^{\mathrm{stat}}_{i,t}(r)$ or by the p-value $\mathrm{Coint}^{p}_{i,t}(r)$. The second is the forecast volatility of the spread, on the reasoning that when a position is opened at an extreme z-score, a more volatile spread is more likely to return toward its mean than to move further away.

$$\gamma^{\mathrm{stat}}_{i,t} = h_{i,t}\,\mathrm{Coint}^{\mathrm{stat}}_{i,t}(r), \qquad \gamma^{p}_{i,t} = \frac{h_{i,t}}{\mathrm{Coint}^{p}_{i,t}(r)}, \tag{9}$$

with $h_{i,t}$ the forecast volatility of spread i, and γ_i denotes the score in use at a given date. Both are recomputed at every date, so the allocation responds to a pair whose cointegration weakens during the testing period.

5.2 Weight Calculation

To allocate more weight to confident pairs without letting any single pair dominate the portfolio, we cap each weight at four times the equally weighted share, $c = \frac{4}{n}$, about 5.5% at $n = 73$. Let A_k be the set of pairs assigned a weight after k passes, starting with $A_0 = \emptyset$, and let $R_k = 1 - \sum_{i \in A_k} w_i$ be the remaining unassigned weight. Pass $k + 1$ assigns

$$w_i = R_k\, c \qquad \text{for every } i \notin A_k \text{ with } \frac{\gamma_i}{\sum_{j \notin A_k} \gamma_j} > R_k\, c, \text{ subject to } \sum_{i=1}^{n} w_i = 1, \tag{10}$$

and the passes repeat until no further pair qualifies. Since R_k falls at every pass, the effective cap tightens as capital is committed. Pairs never assigned receive the smallest assigned weight, and the vector is rescaled to satisfy the constraint.

6 Backtest Results

Three strategies are compared. EW-FE weights the pairs equally and applies the same fixed thresholds to every pair, entering at ± 0.8 and exiting at ∓ 0.05. EW-OE weights the pairs equally and uses the thresholds optimized for profit. OW-OE weights the pairs by equation (10) with the confidence score $\gamma^{p}_{i,t}$ and uses the same thresholds. All three are backtested in Backtrader with market orders filled at the next day's opening price, a transaction cost of 0.1% of traded notional on each side and no slippage. The evaluation window is the 593 trading days from 23 August 2021 to 29 December 2023. The system quality number of Backtrader is $\sqrt{N}$ times the mean per-trade profit and loss divided by its standard deviation, over the N closed trades.

	EW-FE	EW-OE	OW-OE
Total return	158.68%	179.99%	202.22%
Sharpe ratio	1.32	1.17	1.43
Maximum drawdown	28.58%	32.75%	25.59%
System quality number	3.68	2.72	3.28
Number of trades	540	386	393

Table 3: Strategy metrics over the evaluation window, net of transaction costs.

Net of transaction costs, optimizing the thresholds raises the total return from 158.68% to 179.99% but not the risk-adjusted return (Table 3). EW-FE has an annualized volatility of 33.7% and a drawdown of 28.58%, compared to EW-OE's 44.2% and its drawdown of 32.75%. The

Sharpe ratio of EW-OE is therefore 1.17, below the 1.32 of EW-FE. Nevertheless, the confidence weighting, together with optimized thresholds, delivers the best performance. OW-OE earns the highest return, and its volatility of 36.2% and drawdown of 25.59% are well below the 44.2% and 32.75% of EW-OE. The Sharpe ratio is the highest at 1.43, while the number of trades is almost unchanged at 393 against 386 for EW-OE. Relative to EW-OE, focusing on pairs with strong cointegration and volatile spreads improves both risk and return. Figures 4 to 6 plot, from the top panel down, the portfolio value and cash, the per-trade profit and loss and the drawdown behind these results.

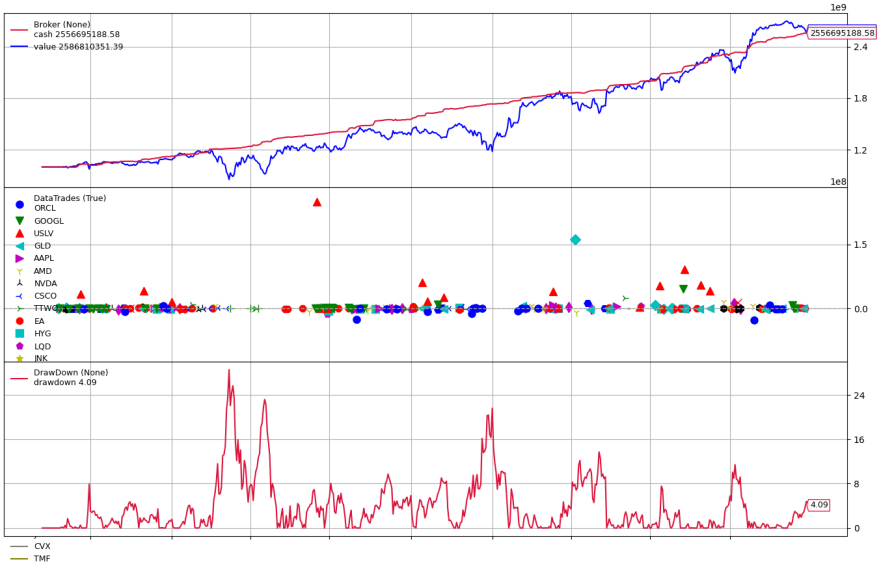


Figure 4: Backtest paths for EW-FE.

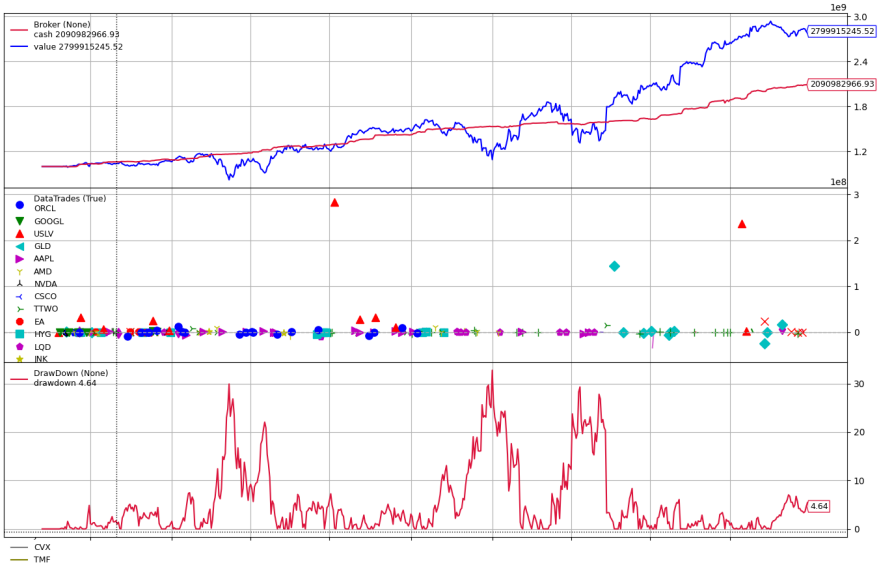


Figure 5: Backtest paths for EW-OE.

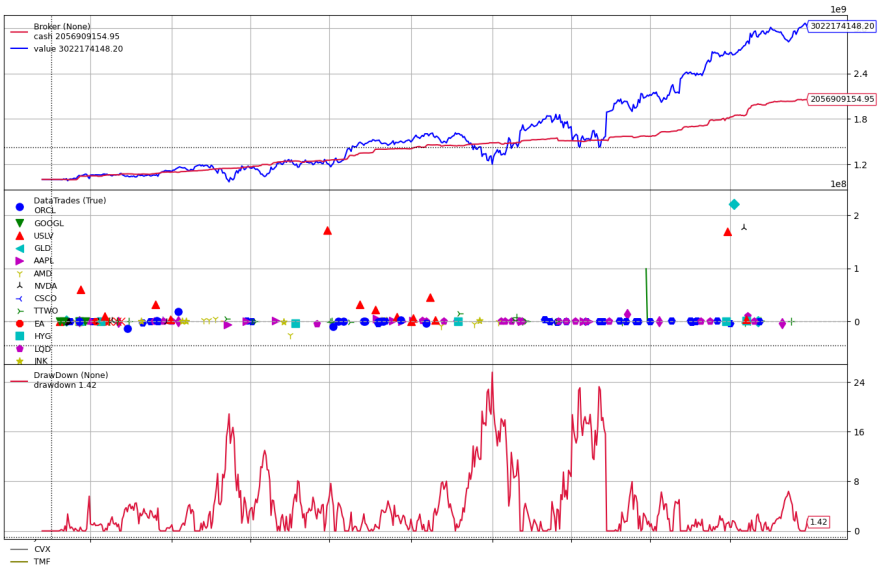


Figure 6: Backtest paths for OW-OE.

7 Conclusion

This project examines whether optimizing the entry and exit thresholds pair by pair and weighting the pairs by a confidence score improve a cointegration-based pairs trading portfolio with DCC-EGARCH spread volatility. Over the 593 trading days of the evaluation window and net of transaction costs, optimizing the thresholds raises the total return from 158.68% to 179.99% but lowers the Sharpe ratio from 1.32 to 1.17. Adding confidence weights based on the rolling cointegration p-value and forecast volatility delivers the highest total return (202.22%), the highest Sharpe ratio (1.43) and the lowest maximum drawdown (25.59%) of the three strategies.

The comparison rests on one evaluation window, assumes no slippage and does not test the differences in Sharpe ratios for significance. Backtesting the Sharpe-ratio thresholds and the confidence score based on the test statistic, both defined above but not traded, is left for future research.

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A Supplementary Figures

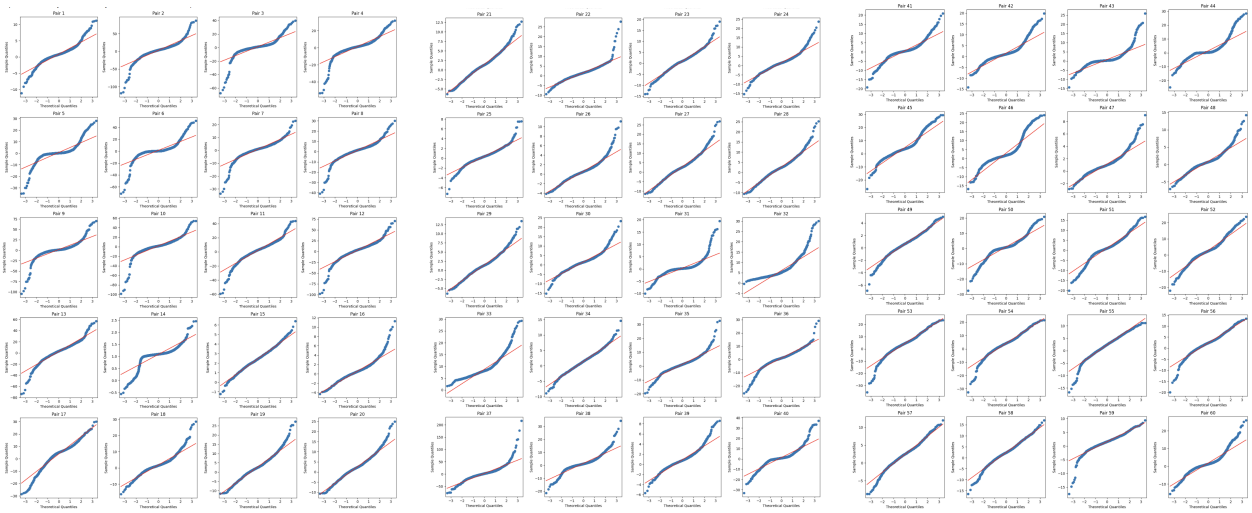


Figure 7: Q-Q plots of the training spreads of Pairs 1–60 against the Gaussian.